ADAPTATION ROUTES LAB / CONCEPT SUMMARY / 04 SEP 2026

When does a learner adapt?

A learner can adapt to a brand-new task by changing its weights from demonstrations, or by changing only its recurrent state.

THE DISTINCTION

Test-time adaptation is not one mechanism. In the optimization route, demonstrations become training signal: the learner reuses or augments examples, computes loss, and takes a backward pass before attempting the unseen task. This is the conceptual shape shown in Panel A. HRM describes a recurrent architecture with high- and low-level modules ([HRM]); TRM describes recursive latent and answer refinement in a tiny network ([TRM]). The toy's animation is not either paper's implementation.

In the context route, the model reads demonstrations in a forward pass and stores useful signal in internal recurrent state while parameters remain fixed for evaluation. BDH-CQ's report describes inference inputs continuously updating recurrent memory ([BDH-CQ §3]). The BDH base paper provides architectural background on recurrent associative state and synaptic plasticity ([BDH]). Panel B visualizes this idea; it does not execute a BDH-CQ checkpoint.

COMPARISON AT A GLANCE

ADAPTATION `optimization / context`	BACKWARD PASS? `toy: yes / no`	EVIDENCE STATUS `mixed`
Training signal changes a solution / recurrent memory carries evidence.	Panel A shows a train-before-predict sequence. Panel B is forward-only at evaluation.	HRM has ARC Prize's approximate independent analysis. BDH-CQ remains developer-reported here.

WHY THE TOY IS USEFUL

The live puzzle is a blue cell moving one step right on a 5×5 grid. Selecting 1–4 demonstrations changes the evidence shared by both panels. Corruption changes one displayed pair and makes the context estimate visibly degrade, while the optimization route runs its toy retraining sequence again. This isolates adaptation structure from benchmark performance.

The failure mode is the point: recurrent state can carry evidence without per-task weight updates, but noisy evidence can interfere with later prediction. The optimization route is not universally immune to bad data; this artifact holds its toy rule constant so the contrast stays legible.

EVIDENCE AND SOURCES

The HRM paper says it uses “two interdependent recurrent modules”; the TRM paper presents recursive latent/answer updates; BDH-CQ says inference inputs update recurrent memory; and the BDH paper supplies background on recurrent associative state. ARC Prize's independent analysis reports that “pre-training task augmentation is critical” and found 300 augmentations sufficient for near-max performance in its experiment. See the exact short quotes and links in SOURCES.md.

`Primary sources: “BDH-CQ: In-Context Learning with Recurrent Latent Reasoning,” arXiv:2608.09888; “The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain,” arXiv:2509.26507; “Hierarchical Reasoning Model,” arXiv:2506.21734; “Less is More: Recursive Reasoning with Tiny Networks,” arXiv:2510.04871; and ARC Prize's independent HRM analysis.`

`Artifact boundary: live toy logic + scripted animation + illustrative state bars. No research checkpoint, gradient computation, or benchmark score is claimed. See README.md and SOURCES.md for methods, licensing, and evidence boundaries.`